ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

DISTANCE LEARNING PROGRAMME
(Academic Session : 2024 - 2025)

JEE (Advanced)
TEST # 04
18-08-2024

JEE(Main+Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Review (Unit Test # 01, 02 & 03)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	B	D	D		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C	A,C	C,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	D	C	A		
SECTION-II	Q.	1	2	3	4	5	6
	A.	4	1	2	5	2	6

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	A	A	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C	A,B,C	B,C			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	A	C	C		
SECTION-II	Q.	1	2	3	4	5	6
	A.	125	7	83	3	4	18

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	A	D	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C,D	A,B	A,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	D	C	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	6	6	4	2	9	6

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (D)

$v = \frac{ds}{dt} = 6t^2 + 6t + 2 \Rightarrow a = \frac{dv}{dt} = 12t + 6 = 0$

$\Rightarrow t = -\frac{1}{2}$ which is impossible. Therefore acceleration will not be zero at any time

3. Ans (D)

Component in x-y plane = $20 \cos 30°$

Component on z axis = $20 \sin 30°$

so

$\vec{A} = 20 \cos 30° \cos 60° \hat{i} + 20 \cos 30° \sin 60° \hat{j} + 20 \sin 30° \hat{k}$

$\vec{A} = 5\sqrt{3}\hat{i} + 15\hat{j} + 10\hat{k}$

4. Ans (D)

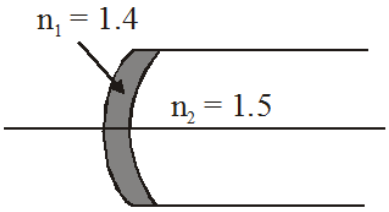
$-11 = -5t + \frac{1}{2} at^2$

For More Material Join: @JEEAdvanced_2025

PART-1 : PHYSICS

SECTION-I (ii)

6. Ans (A,C)



When rays are moving from air to glass,

$$\frac{1.5}{f_1} = \frac{(1.4-1)}{+R} + \frac{(1.5-1.4)}{+R}$$

$$\frac{1.5}{f_1} = \frac{0.4}{R} + \frac{0.1}{R} = \frac{0.5}{R}$$

$|f_1| = 3R$

When rays are moving from glass to air,

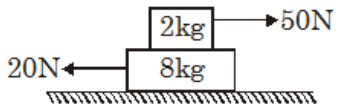
$$\frac{1}{F_2} = \frac{(1-1.4)}{-R} + \frac{(1.4-1.5)}{-R} = \frac{0.5}{R}$$

$|f_2| = 2R$

PART-1 : PHYSICS

SECTION-I (iii)

8. Ans (D)



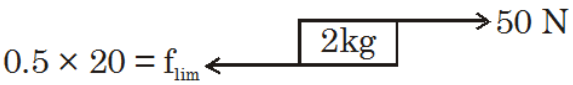
Min force required to move 8 kg block on surface = μN

= 0.4 (100) = 40 N

so 8 kg block will remain at rest

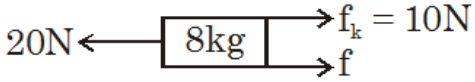
$\boxed{a_{8kg} = 0}$

FBD of 2 kg



$$a_{2kg} = \frac{50-10}{2} = 20\ m/s^2$$

(iv) friction between 8 kg block & ground will be static



so $f_{static} = 10$ N

10. Ans (C)

gravitational potential is $\alpha \left(\frac{-1}{r}\right)$ so any position far away has more potential. Gravitational field inside a uniform hollow sphere is zero due to its own mass, but it is non-zero due to other masses present near by.

11. Ans (A)

$\vec{v} = 6\hat{i} + (20-4t)\hat{j}$

time when $v_y = 0$

$20 - 4t = 0$

$t = 5$

$T = 2 \times t = 10$ sec

$$\tan 53 = \frac{20-4t}{6}$$

$$\frac{4}{3} = \frac{20-4t}{6}$$

$t = 3$ sec

$R = v_x \times T$

$= 6 \times 10 = 60$ m

$$H = \frac{v_y^2}{2 \times a} = \frac{20 \times 20}{2 \times 4} = \frac{400}{8} = \frac{200}{4} = 50m$$

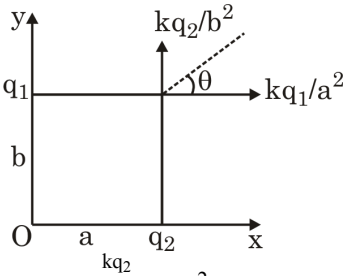
PART-1 : PHYSICS

SECTION-II

1. Ans (4)

$$a_E = \frac{GM_S}{r_E^2} \text{ and } a_S = \frac{GM_S}{\left(\frac{r_E}{2}\right)^2} \Rightarrow a_S = 4a_E$$

3. Ans (2)

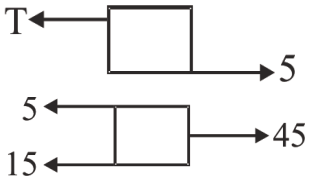


$$\tan\theta = \frac{\frac{kq_2}{b^2}}{\frac{kq_1}{a^2}} = \frac{a^2}{b^2}\frac{q_2}{q_1} = \frac{4}{1} \times \frac{1}{2} = 2$$

For More Material Join: @JEEAdvanced_2025

4. **Ans (5)**

$5 - T = 0 \Rightarrow T = 5$



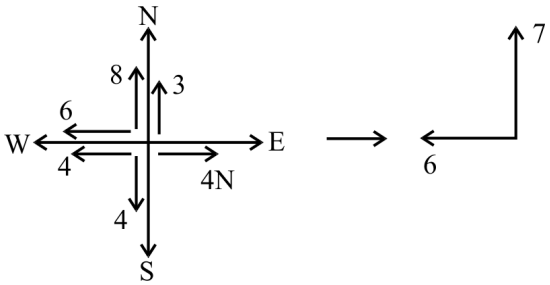
5. **Ans (2)**

$[E] = ML^2T^{-2}$

$$n = 5\left(\frac{M_1}{M_2}\right)\left(\frac{L_1}{L_2}\right)^2\left(\frac{T_1}{T_2}\right)^{-2}$$

$$= 5\,(2)\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^{-2}$$

6. **Ans (6)**

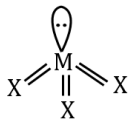


To have resultant along north, force towards west should be cancelled.

PART-2 : CHEMISTRY

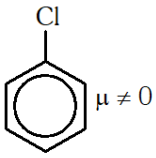
SECTION-I (i)

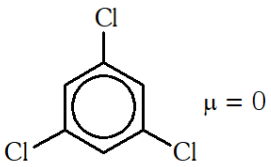
1. **Ans (D)**

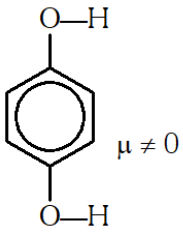


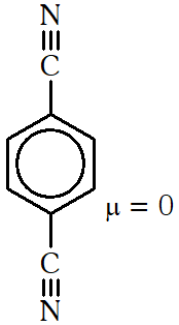
X used its
P orbital for π bonds
Formation & hybridization is sp^3

2. **Ans (A)**









3. **Ans (A)**

such type of question is solved by excess product

$$2SO_3(g) \xrightarrow{50\%} 2SO_2(g) + O_2(g)$$

	6400 g	6400 g
mol	$\frac{6400}{64}$	$\frac{6400}{32}$
	100	200
		excess

Moles of SO_3 wrt 200 moles of O_2 = 400 mol

mass = 400 × 80 = 3200

= 32 kg (theoretical)

actual mass of $SO_3 = 32 \times \frac{100}{50} = 64$kg

4. **Ans (C)**

Each Na^+ ion is closely surrounded by 6 other Cl^- ions

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,C)**

Thermal stabilits ⇒ LiOH < KOH < RbOH

Bond length ⇒ HF < HCl < HBr < HI

6. **Ans (A,B,C)**

B.E : $N_2 > C_2 > O2 > B2 > F2$

(A) Interelectronic repulsion

(B) Based on B.O and atomic size

(D) HgS > CdS > ZnS : Poor shielding by d and f-orbital

For More Material Join: @JEEAdvanced_2025

7. **Ans (B,C)**

By $PV = nRT$

At "A"

$2 \times 8 = 2 \times R \times T_A$

$T_A = \frac{8}{R}$

Between A & B , V is constant

$\frac{P_A}{T_A} = \frac{P_B}{T_B}$

$\frac{2}{8/R} = \frac{P_B}{300} \Rightarrow P_B = 75R$

Between A & D ,P is constant

$\frac{V_A}{T_A} = \frac{V_D}{T_D}$

$\frac{8}{8/R} = \frac{10}{T_D} \Rightarrow T_D = \frac{10}{R}$

PART-2 : CHEMISTRY

SECTION-I (iii)

8. **Ans (D)**

NF_3 is polar (pyramidal shape)

XeF_5^+ shape is sq. pyramidal (polar),

bond angle < 90°

ICl_2^+ shape is bent (polar),

bond angle < 109°28'

11. **Ans (C)**

(P) 20 gm of KOH present in 100 gm solution

$V = \frac{100}{1.02} = 98.04$ ml

$M = \frac{20/56}{98.04} \times 1000 = 3.64$ M

(Q) 444 gm $CaCl_2$ in 1 lit. solute

$\frac{444}{111/1} = \frac{444}{111} = 4$ M

(R) 1.204×10^{24} molecules of water = 2 mole

2 mole of water at 4°C is 36 gm or 36 ml

(S) Volume of 0.2 M NaOH solution

$0.2 \times V = 1 \times 10^{-3}$ mole

$V = \frac{1 \times 10^{-3}}{0.2} \times 1000$

$V = 5$ ml

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (125)**

Possible subshells are present in the 8th period:

8s 7d 6f 5g 8p

Total electrons = 5 + 25 + 35 + 45 + 15 = 125

2. **Ans (7)**

SF_6, ClF_6^+, $BeCl_2$, SO_3, PH_4^+, OF_2, ICl_2^+

3. **Ans (83)**

Moles of $CuCl_2 = 10 \times \frac{1}{2} \times \frac{1}{0.6} \times 2 \times \frac{1}{0.5} \times \frac{1}{0.4}$

$\Rightarrow 83.33$ moles

Balanced equation :

$CuCl_2 + 2KCN \xrightarrow{40\%} Cu(CN)_2 + 2KCl$

$2Cu(CN)_2 \xrightarrow{50\%} Cu_2(CN)_2 + (CN)_2 \uparrow$

$Cu_2(CN)_2 + 6KCN \xrightarrow{60\%} 2K_3[Cu(CN)_4]$

4. **Ans (3)**

$\Delta G = \Delta G^\circ + RT \ln Q$

$\Delta G^\circ = 2\Delta G_f^\circ(NO_2) - \Delta Gf^\circ(N_2O_4) = 0$

$\Delta G = 8 \times 300 \ln\left(\frac{2^2}{1}\right) = 3360$ J

5. **Ans (4)**

$\frac{1}{P_{Total}} = \frac{Y_A}{P_A^o} + \frac{Y_B}{P_B^o}$

$\frac{1}{99} = \frac{1 - Y_B}{100} + \frac{Y_B}{80}$

$\frac{1}{99} = \frac{4 - 4Y_B + 5Y_B}{400}$

$400 = 396 + 99Y_B$

$Y_B = \frac{4}{99} \times 100 \simeq 4\%$

For More Material Join: @JEEAdvanced_2025

6. **Ans (18)**

1mL ⟶ 1.8 gm 1000 mL ⟶ 1800 gm.

Mass of $H_3PO_4 \Rightarrow \frac{98}{100} \times 1800 = 98 \times 18$

$$\text{Molarity} = \frac{10 \times 1.8 \times 98}{98} = 18 \text{ M}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

Expression

$= (\alpha\tan\gamma + \beta\cot\gamma)(\alpha\cot\gamma + \beta\tan\gamma)$

$- 4\alpha\beta\cot^2 2\gamma$

$= \alpha^2 + (\tan^2\gamma + \cot^2\gamma)\,\alpha\beta + \beta^2$

$- 4\alpha\beta \dfrac{\cos^2 2\gamma}{\sin^2 2\gamma}$

$= \alpha^2 + \beta^2 + \alpha\beta$

$$\left[\left(\frac{\sin^2\gamma}{\cos^2\gamma} + \frac{\cos^2\gamma}{\sin^2\gamma}\right) - \frac{4(\cos^2\gamma - \sin^2\gamma)^2}{4\sin^2\gamma\cos^2\gamma}\right]$$

$= \alpha^2 + \beta^2 + \alpha\beta$

$$\left[\frac{(\sin^4\gamma + \cos^4\gamma) - (\cos^4\gamma + \sin^4\gamma - 2\sin^2\gamma\cos^2\gamma)}{\sin^2\gamma\,\cos^2\gamma}\right]$$

$= \alpha^2 + \beta^2 + 2\alpha\beta = (\alpha + \beta)^2$,

Which is independent of γ and dependent on α, β.

$\therefore$ options (D) is correct.

2. **Ans (A)**

$P(x = 4t^2 + 3, y = 8t^3 - 1)$

let $Q(4t_1^2 + 3, 8t_1^3 - 1)$

at P, $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt} = \dfrac{24t^2}{8t} = 3t$

$\therefore$ tangent at P is $y - 8t^3 + 1 = 3t\,(x - 4t^2 - 3)$

Q will satisfy it

$\therefore 8t_1^3 - 8t^3 = 3t\,(4t_1^2 - 4t^2)$

$8(t_1 - t)(t_1^2 + t_1 t + t^2) = 3t.\ 4(t_1 - t)\,(t_1 + t)$

$2(t_1^2 + t_1 t + t^2) = 3t(t_1 + t)$

$2t_1^2 + 2t_1 t + 2t^2 = 3t\,t_1 + 3t^2$

$2t_1^2 - t_1 t - t^2 = 0$

$(t_1 - t)\,(2t_1 + t) = 0$

$t_1 = -\dfrac{t}{2}$

$\therefore Q\,(t^2 + 3,\ -t^3 - 1)$

3. **Ans (D)**

$f(x) = -2x^3 - 9x^2 - 12x + 1$

$\Rightarrow f'(x) = -6x^2 - 18x - 12$

To be decreasing

$f'(x) < 0$, i.e., $-6x^2 - 18x - 12 < 0$

$\Rightarrow x^2 + 3x + 2 > 0$

$\Rightarrow (x + 2)(x + 1) > 0$

Therefore either $x < -2$ or $x > -1$

$\Rightarrow x \in (-1,\ \infty)$ or $(-\infty, -2)$

4. **Ans (C)**

$$\lim_{x\to 0} \frac{\dfrac{(-2)(2x - x^2)^2}{2}}{\left(-\dfrac{x^2}{2}\right)} = 8$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,C,D)**

$f(x) =$

$$\begin{cases} 1 & x \in \left(2n\pi, 2n\pi + \frac{\pi}{2}\right) & n \in I^+ \cup \{0\} \\ \frac{1+5\log_e x}{3} & x = (4n+1)\frac{\pi}{2} & n \in I^+ \cup \{0\} \\ \frac{5\log_e x}{2} & x \in \left(2n\pi + \frac{\pi}{2}, 2n\pi + \frac{3\pi}{2}\right) & n \in I^+ \cup \{0\} \\ \frac{1+5\log_e x}{3} & x \in 2n\pi + \frac{3\pi}{2} & n \in I^+ \cup \{0\} \\ 1 & x \in \left(2n\pi + \frac{3\pi}{2}, 2n\pi + 2\pi\right) & n \in I^+ \cup \{0\} \end{cases}$$

$f(x)$ is discontinuous at all positive odd integral multiple of $\dfrac{\pi}{2}$

$f(x)$ is continuous at integral multiple of π

6. **Ans (A,B)**

$$\left(\frac{x^2 + x + 2}{x^2 + x + 1}\right)^2 - (a - 3)\left(\frac{x^2 + x + 2}{x^2 + x + 1}\right) + (a - 4) = 0$$

$$y = \frac{x^2 + x + 2}{x^2 + x + 1} = 1 + \frac{1}{x^2 + x + 1} \in \left(1, \frac{7}{3}\right]$$

$y^2 - (a - 3)\,y + (a - 4) = 0$

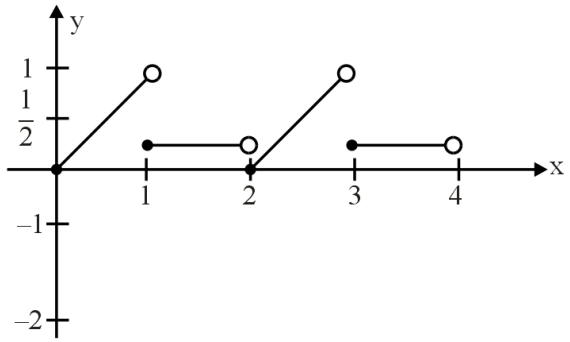
$\Rightarrow (y^2 + 3y - 4) - a\,(y - 1) = 0$

$\Rightarrow (y - 1)\,(y + 4 - a) = 0 \Rightarrow 1 < a - 4 < \dfrac{7}{3}$

$a \in \left(5, \dfrac{19}{3}\right]$

For More Material Join: @JEEAdvanced_2025

7. Ans (A,D)



$$f(x)=\begin{cases} x & , \ 0 \leqslant x < 1 \\ \frac{1}{2} & , \ 1 \leqslant x < 2 \\ x-2 & , \ 2 \leqslant x < 3 \\ \frac{1}{2} & , \ 3 \leqslant x < 4 \end{cases}$$

PART-3 : MATHEMATICS

SECTION-I (iii)

8. Ans (D)

(P) $2\ell n(2^x-5)=\ell n\left(2\left(2^x-\frac{7}{2}\right)\right)$

Let $2^x=t$

$t^2-12t+32=0 \quad \Rightarrow \quad t=4$ or $t=8$

$\therefore \quad x=2$ (rejected) or $x=3$

(Q) Take log both sides to base '10'

$\log_{10}x\left(\log_{10}\left(\frac{5x}{2}\right)-1\right)=0$

$x=1$ or $x=4$

Hence, sum = 5

(R) $\alpha=2^{\log_2(\log_3 5)}=\log_3 5$

$\beta^{\alpha}=9 \ \Rightarrow \ \beta^{\log_3 5}=9$

$\frac{1}{5}\left(\beta^{\alpha^2}\right)=\frac{1}{5}(\beta^{\alpha})^{\alpha}=\frac{1}{5}(9)^{\log_3 5}$

$=\frac{1}{3}(5)^{\log_3 9}=\frac{25}{5}=5$

(S) $\sqrt{6+\sqrt{6+\sqrt{6+\ldots\infty}}}=3$

$\log_{\sqrt{3}}(3)=(2)$

9. Ans (D)

(P) $81^{\sin^2 x}+\frac{81}{81^{\sin^2 x}}=30$

Let $81^{\sin^2 x}=t$

$t+\frac{81}{t}=30$

$t=27$ or $t=3$

$81^{\sin^2 x}=27$

$3^{4\sin^2 x}=3^3$ & $3^{4\sin^2 x}=3^1$

$\sin x=\frac{\sqrt{3}}{2}$ & $\sin x=\frac{1}{2}$

$x=\frac{\pi}{3}$ or $x=\frac{\pi}{6}$

(Q) $x+1=3$ and $y+2=3$

$x=2$ and $y=1$

$x+y=3$

(R) $2\times 2\times 3=$ min value

min. value = 12

(S) $(1-\alpha)(1-\beta)=1+(\sin\theta+\cos\theta)$

$\therefore$ maximum value $=1+\sqrt{2}$

10. Ans (C)

(P) $\lim\limits_{x\to 0}\frac{a(\cos 3x)+(1-\cos 2x)}{x+\sin x+\cos x}=1$

$\lim\limits_{x\to 0} a(\cos 3x)+(1-\cos 2x)=\lim\limits_{a\to 0} x+\sin x+\cos x$

(Q) $f(x)=(a-7)\log 3x+(b-2)ex+\sin x$ for

function to be bounded a = 7; b = 2

(R) $f(x)=x^2+px+10$; for $f(x)$ to be even $p=0$

$\lim\limits_{x\to 0}\frac{1-\cos(1-\cos x)}{x^4}=\frac{m}{n}$ then $m+n=9$

(S) Point is are x = 2,3

For More Material Join: @JEEAdvanced_2025

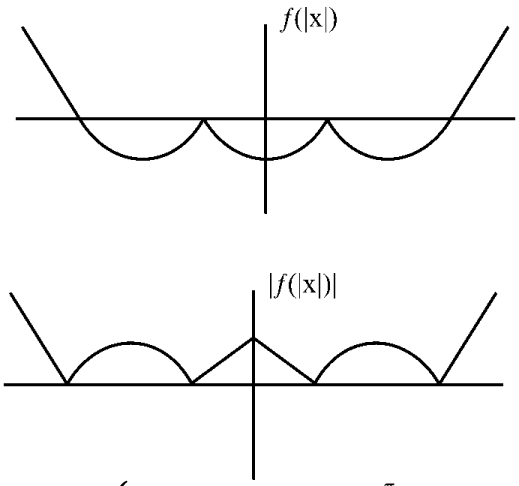
11. Ans (B)

(P) $f\left(\frac{\pi}{2}+h\right)=\lim\limits_{h\to0}\frac{\left(e^{\tan\left(\frac{\pi}{2}+h\right)}-1\right)}{\left(e^{\tan\left(\frac{\pi}{2}+h\right)}+1\right)}=-1$

$f\left(\frac{\pi}{2}-h\right)=\lim\limits_{h\to0}\frac{e^{\tan\left(\frac{\pi}{2}-h\right)}-1}{\left(e^{\tan\left(\frac{\pi}{2}-h\right)}+1\right)}=1$

Jump = 2

(Q)



(R) $f(x)=\begin{cases} x & ; \quad 0<x<\frac{\pi}{4} \\ b & ; \quad x=0 \\ \sin x & ; \quad -\frac{\pi}{4}<x<0 \end{cases}$

$f(0)=f(0+h)=f(0-h)\Rightarrow b=0$

$f(x)=\begin{cases} x & ; \quad 0\leqslant x<\frac{\pi}{4} \\ \sin x & ; \quad -\frac{\pi}{4}<x<0 \end{cases}$

$f'(0^+)=f'(0^-)=1$

(S) $\ell=\frac{1-\cos 9x}{(e^x-1)^2}=\frac{81}{2}$

$\left[\frac{\ell}{9}\right]=[4.5]=4$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (6)

$\because$ (x, 1), (y, 2), (z, 1), (2, 3) $\in$ A × B

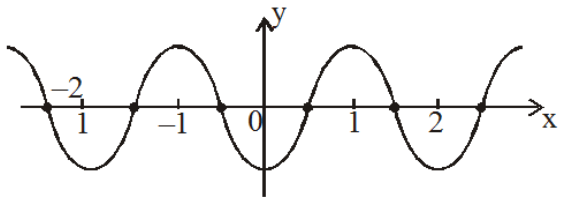
$\Rightarrow$ A = {x, y, z, 2}

B = {1, 2, 3}

Number of onto map = ${}^4C_2 3! = 36$

2. Ans (6)

$\lim\limits_{x\to\infty}P(x)\longrightarrow\infty$ and $\lim\limits_{x\to(-\infty)}P(x)\longrightarrow\infty$.



P(0) = −1 and P(x) is even this P(1) = P(−1) = 1

& P(2) = P(−2) = −1.

Hence, from IVT minimum no. of zeroes are 6.

3. Ans (4)

Required limit is

$\frac{d}{dx}\left(h^{-1}(x)\right)\Big|_{x=1}$

$\because f^{-1}(x)=x-1;\ g^{-1}(x)=x^3+x+1$

Also h(x) = g(f(x))

$\Rightarrow h^{-1}(x)=f^{-1}(g^{-1}(x))=x^3+x$

$\Rightarrow \frac{d}{dx}\left(h^{-1}(x)\right)=3x^2+1$

For More Material Join: @JEEAdvanced_2025

4. **Ans (2)**

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \sum_{k=1}^{\infty} \frac{k}{2^k}$$

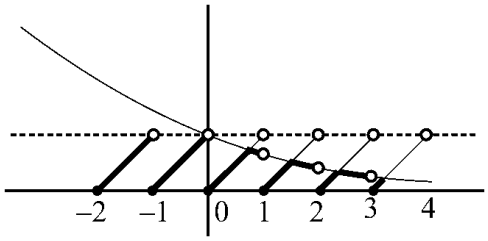
$$S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \ldots.$$

$$\frac{S}{2} = \frac{1}{2^2} + \frac{2}{2^3} + \ldots. = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \ldots..$$

$$\frac{S}{2} = 1 \Rightarrow S = 2$$

$$2\sum_{n=1}^{\infty} \frac{1}{2^n} = 2\left(\frac{1}{2} + \frac{1}{2^2} + \ldots.....\right) = 2$$

5. **Ans (9)**



6. **Ans (6)**

$$f(x + y) = f(x) + f(y) + x^2y + xy^2$$

$$f'(x) = \lim_{h \to 0} \frac{f(x) + f(h) + x^2h + xh^2 - f(x)}{h}$$

$$= x^2 + \lim_{h \to 0} \frac{f(h)}{h}$$

$$f'(x) = x^2 + 2$$

$$f(x) = \frac{x^3}{3} + 2x + c$$

For More Material Join: @JEEAdvanced_2025